AIRAVATA · CODE-REVIEWED EXPLANATION

Meta WhatsApp Onboarding and Airavata Credits

Why onboarding can be blocked

Meta business verification and Meta payment-card approval are separate checks.

AutoGamma's business can be fully verified while Meta still rejects the payment card because of:

- Bank or country restrictions
- International or recurring payments being disabled
- 3-D Secure or payment authentication failure
- A prepaid, virtual, or unsupported card type
- A billing address or business-detail mismatch
- An existing unpaid Meta balance
- A payment account or Business Manager restriction
- A currency or billing-country mismatch
- Meta requiring a payment method for WhatsApp Business messaging limits

> **Key point:** Airavata cannot bypass this validation because it happens inside Meta's Embedded Signup and billing system. If Meta rejects the card before signup completes, the process usually does not reach the successful `FINISH` event, so Airavata does not receive the WABA and phone-number details.

What happens after Meta accepts the card

1. AutoGamma logs into its Airavata account.
2. AutoGamma selects **Connect Facebook** or **Embedded Signup**.
3. Meta handles business selection, WABA creation or selection, phone-number selection, payment setup, and permissions.
4. Meta sends Airavata an authorization code.
5. Airavata exchanges the code with Meta.
6. Airavata receives the WABA ID, Phone Number ID, and WhatsApp access token.
7. Airavata encrypts and stores the access token against AutoGamma's user account.
8. The account is marked connected only after the encrypted credentials are saved successfully.

> The connection is tenant-specific, so AutoGamma's WhatsApp credentials are not mixed with another client's credentials.

Which account sends the messages?

After onboarding, ordinary client accounts send messages through the credentials stored for that specific client:

- AutoGamma's WABA
- AutoGamma's Phone Number ID
- AutoGamma's Meta access token

The webhook also identifies the owning client from the Meta Phone Number ID, allowing incoming messages and delivery updates to be associated with the correct client workspace.

How Admin-assigned credits work

- AutoGamma's internal credit balance increases.
- A credit transaction is recorded.
- Template messages check the available balance before sending.
- Credits are reserved before Airavata calls Meta.
- If Meta rejects the message, the reserved credits are refunded.
- If the message succeeds, a deduction transaction is recorded.

Current credit rules

Message type	Airavata credit behavior
Marketing template	Uses the configured Marketing rate
Utility template	Uses the configured Utility rate
Authentication template	Uses the configured Authentication rate
Normal free-form session message	Not charged Airavata credits

Important distinction: Meta billing vs Airavata credits

Airavata credits and Meta billing are separate systems.

Airavata credits control usage inside the Airavata platform. They do not replace Meta's requirement for an accepted payment method.

Therefore AutoGamma may need both:

1. An accepted payment method in Meta for WhatsApp Business billing and account activation requirements.
2. Credits assigned by Master Admin for Airavata's internal message-credit rules.

The Meta card is used for Meta's own billing and account rules. The credits assigned in Airavata are used for Airavata's platform controls.

How to locate the failure

1. The Embedded Signup window reaches Meta's successful completion step.
2. Browser logs show the Embedded Signup `FINISH` event.
3. The API logs a successful Meta authorization-code exchange.
4. The API logs that encrypted WhatsApp credentials were stored.
5. AutoGamma's account shows Connected status, a WABA ID, and a Phone Number ID.

6. A test message is sent from AutoGamma's WhatsApp number, not the ecosystem or administrator number.

> If Meta rejects the card before the `FINISH` event, the failure is on Meta's billing or payment side. No WABA credentials will be stored by Airavata until the Meta onboarding flow completes successfully.

Logging and failure checkpoints

The system has browser-console logs, API workflow logs, and user-facing error messages for the main onboarding and messaging paths.

Checkpoint coverage

Checkpoint	What the system provides
Embedded Signup starts	Browser console logs the signup configuration and start
Meta sends signup events	Browser console logs `WA_EMBEDDED_SIGNUP` events
Signup completes	Browser console logs `FINISH IDs captured` with WABA and Phone Number IDs
Authorization code is sent	Browser console logs `POST /whatsapp/onboard`
Meta token exchange	API logs the response status and whether the token, WABA, and phone ID were received
Meta token exchange fails	API logs and returns Meta's error message
Credential encryption	API logs encryption start and encryption failures
Credential storage	API logs successful encrypted storage or MongoDB save failures
User connection state	API stores connected status, WABA ID, and Phone Number ID
Test message fails	Meta status, code, subcode, type, message, and Facebook trace ID are logged
User-facing failure	The frontend displays the backend or Meta error in a toast

Important success logs

Browser logs should include:

- `[WhatsApp Embedded Signup] FINISH IDs captured`
- `[WhatsApp Embedded Signup] Sending code to backend`
- `POST /whatsapp/onboard`

API logs for successful onboarding should include:

- `WhatsApp Embedded Signup: starting Meta code exchange`
- `WhatsApp Embedded Signup: Meta token exchange response received`
- `WhatsApp credentials encrypted and stored in whatsappcredentials`
- `WhatsApp Business Account connected via Embedded Signup`

Important failure logs

The API reports specific failures such as:

- `Meta token exchange failed`
- `Meta phone number lookup failed`
- `WhatsApp credential encryption failed`
- `WhatsApp credential MongoDB save failed`
- `Meta onboarding did not return a WABA and phone number`
- `WhatsApp credentials could not be saved to the account`

For a connected AutoGamma account, the connection status should show:

- `connected: true`
- A WABA ID
- AutoGamma's Phone Number ID
- `credentialStored: true`
- `credentialReadable: true`

Limitation when Meta rejects the card

If Meta rejects the card inside the Embedded Signup window before the `FINISH` event, Airavata may not receive an authorization code or an API request. In that situation, no WABA or Phone Number ID is stored because the failure happens inside Meta before Airavata onboarding begins.

The logs are available in the browser developer console and API workflow logs. They are not currently stored as a permanent onboarding audit history inside the application. Also, if Facebook returns a connected response without an authorization code, the frontend may stop without showing a detailed error toast.

Conclusion

The card rejection is not caused by AutoGamma's business-verification status and is not something Airavata can override. Once Meta accepts the payment method and Embedded Signup completes, the existing Airavata design stores AutoGamma's connection separately, sends through AutoGamma's business account, and applies the credits assigned by Master Admin according to the configured message category rates.

Prepared from a review of the current Airavata onboarding, credential storage, messaging, webhook, and credit-deduction flow.